

Development Plan

Software Engineering

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Table 1: Revision History

Date	Developer(s)	Change
Date1	Name(s)	Description of changes
Date2	Name(s)	Description of changes
...	...	...

1 Introduction

This document outlines the proposed development plan for the team. Sections 1-3 contain legal and regulatory information, sections 4-6 contain logistical details for team communication and roles, and Section 7-10 contain timeline and technology logistics. The team charter is also included at the end of this document.

2 Confidential Information?

The images used to train the machine learning model are confidential and cannot be shared publicly. The images will be provided by various municipalities and are subject to a confidentiality agreement. The team will not share any images or data that is considered confidential. A confidentiality agreement has not been completed yet, but a template is available via this link: [Capstone Template](#).

3 IP to Protect

For this project, there is IP to protect. An IP agreement has not been completed yet, but a template is available via this link: [Capstone Template](#).

4 Copyright License

The team is explicitly choosing not to include a copyright license for this project, meaning that the code is under exclusive copyright. This information is also available in the license file: [LICENSE](#).

5 Team Meeting Plan

Team members will meet on a weekly basis on Tuesdays from 11 AM - 12:30 PM. Meetings will be located in-person at one of McMaster's libraries. Team members are expected to come prepared by having reviewed the current state of the deliverable at that time. Meetings may occur more than once per week if needed, which will be held virtually on Discord. Supervisor meetings will be held regularly on a biweekly basis, but the team will be open to meeting more frequently if the supervisor requests it. All members will take turns chairing meetings and taking meeting notes. Meetings should be structured as follows.

At the start of every meeting, the chair will go over the purpose of the meeting and discuss what is on the agenda.

After outlining the agenda, the chair will go through each meeting item one-by-one. Each item should have a person responsible for that task to report on

progress, expected outcomes, and explain next steps if necessary. Each item should have a clear time allocation, and expected deadline.

At the end of the meeting, the chair will summarize all key decisions made, and restates actions items for the

[How often will the team meet? where? when? —SS]

[If the meeting is a physical location (not virtual), out of an abundance of caution for safety reasons you shouldn't put the location online —SS]

[To help schedule meetings, you can use the lecture/tutorial time when we do not have anything else scheduled. —SS]

[How often will you meet with your supervisor/advisor? when? where? —SS]

[Will meetings be virtual? At least some meetings should likely be in-person. —SS]

[How will the meetings be structured? There should be a chair for all meetings. There should be an agenda for all meetings. —SS]

6 Team Communication Plan

[Issues on GitHub should be part of your communication plan. —SS]

7 Team Member Roles

[You should identify the types of roles you anticipate, like notetaker, leader, meeting chair, reviewer. Assigning specific people to those roles is not necessary at this stage. In a student team the role of the individuals will likely change throughout the year. —SS]

8 Workflow Plan

- How will you be using git, including branches, pull request, etc.?
- How will you be managing issues, including template issues, issue classification, etc.?
- Use of CI/CD

9 Project Decomposition and Scheduling

- How will you be using GitHub projects?
- Include a link to your GitHub project

[How will the project be scheduled? This is the big picture schedule, not details. You will need to reproduce information that is in the course outline for deadlines. —SS]

10 Proof of Concept Demonstration Plan

The main risk towards the success of the project is the ability to accurately detect and classify the state of the road based on the information given. During the proof of concept demonstration, we will demonstrate that our model can accurately detect and classify the state of the road based on the information given.

11 Expected Technology

[What programming language or languages do you expect to use? What external libraries? What frameworks? What technologies. Are there major components of the implementation that you expect you will implement, despite the existence of libraries that provide the required functionality. For projects with machine learning, will you use pre-trained models, or be training your own model? —SS]

[The implementation decisions can, and likely will, change over the course of the project. The initial documentation should be written in an abstract way; it should be agnostic of the implementation choices, unless the implementation choices are project constraints. However, recording our initial thoughts on implementation helps understand the challenge level and feasibility of a project. It may also help with early identification of areas where project members will need to augment their training. —SS]

Topics to discuss include the following:

- Specific programming language
- Specific libraries
- Pre-trained models
- Specific linter tool (if appropriate)
- Specific unit testing framework
- Investigation of code coverage measuring tools
- Specific plans for Continuous Integration (CI), or an explanation that CI is not being done
- Specific performance measuring tools (like Valgrind), if appropriate
- Tools you will likely be using?

Python will be the primary programming language used for this project. We will be using libraries such as OpenCV for image processing, TensorFlow or PyTorch for machine learning, and NumPy for numerical computations. We will also utilize pre-trained models for road state detection and classification.

Ruff will be used as the linter for Python code, and PyTest will be used as the unit testing framework. We will investigate code coverage measuring tools such as Coverage.py to ensure that our tests are comprehensive.

For version control, we will use Git and GitHub. We will use CI for linting and unit test passing. Since we are using a Github repo, Github Actions is the preferred method of choice for CI.

12 Use of AI and LLMs

[This section is a continuation of the previous section. The role of LLMs in your project as important enough to warrant their own section. How will your team be using LLMs for code and documentation development? What tools will you be using? How will you verify the output of your LLMs? —SS]

13 Coding Standard

PEP 8 will be used as the coding standard for Python code.

14 Appendix — Generative AI Use Disclosure

[ChatGPT (version 5) was used to brainstorm ideas for the template for this disclose section. —SS]

[This section is about the use of AI to write this document, not your planned use of AI for your project. You discuss that topic in one of the sections in the body of this report. —SS]

14.1 AI Tools Used

[Give the name of the AI tool(s) used and the version. —SS]

14.2 How and Where Used

[Explain what the tool(s) were used for. Examples include: brainstorming, explaining a technical concept, reviewing the document, generating or modifying text, generating or modifying diagrams, identifying errors or omissions or inconsistencies. —SS]

[For each of the uses, identify which parts of the document were affected. —SS]

[You do not need to report every interaction, but you should disclose the uses that materially contributed to your work —SS]

14.3 Verification of AI-Generated Content

[

- Describe how the team checked AI-generated or AI-assisted material before including it in the document.
- In particular, verify factual claims, technical assertions, requirements, calculations, references, and other information for which an AI system may produce incorrect or fabricated results.
- **The team is responsible for the accuracy, completeness, consistency, and appropriateness of the submitted document, regardless of whether material was generated by a human or an AI system.**

—SS]

Appendix — Reflection

[Not required for CAS 741 —SS]

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. Why is it important to create a development plan prior to starting the project?
2. In your opinion, what are the advantages and disadvantages of using CI/CD?
3. What disagreements did your group have in this deliverable, if any, and how did you resolve them?

Appendix — Team Charter

[borrows from University of Portland Team Charter —SS]

External Goals

[What are your team’s external goals for this project? These are not the goals related to the functionality or quality of the project. These are the goals on what the team wishes to achieve with the project. Potential goals are to win a prize at the Capstone EXPO, or to have something to talk about in interviews, or to get an A+, etc. —SS]

Attendance

Expectations

[What are your team’s expectations regarding meeting attendance (being on time, leaving early, missing meetings, etc.)? —SS]

Acceptable Excuse

[What constitutes an acceptable excuse for missing a meeting or a deadline? What types of excuses will not be considered acceptable? —SS]

In Case of Emergency

[What process will team members follow if they have an emergency and cannot attend a team meeting or complete their individual work promised for a team deliverable? —SS]

Accountability and Teamwork

Quality

[What are your team’s expectations regarding the quality of team members’ preparation for team meetings and the quality of the deliverables that members bring to the team? —SS]

Attitude

[What are your team’s expectations regarding team members’ ideas, interactions with the team, cooperation, attitudes, and anything else regarding team member contributions? Do you want to introduce a code of conduct? Do you want a conflict resolution plan? Can adopt existing codes of conduct. —SS]

Stay on Track

[What methods will be used to keep the team on track? How will your team ensure that members contribute as expected to the team and that the team performs as expected? How will your team reward members who do well and manage members whose performance is below expectations? What are the consequences for someone not contributing their fair share? —SS]

[You may wish to use the project management metrics collected for the TA and instructor for this. —SS]

[You can set target metrics for attendance, commits, etc. What are the consequences if someone doesn't hit their targets? Do they need to bring the coffee to the next team meeting? Does the team need to make an appointment with their TA, or the instructor? Are there incentives for reaching targets early? —SS]

Team Building

[How will you build team cohesion (fun time, group rituals, etc.)? —SS]

Decision Making

[How will you make decisions in your group? Consensus? Vote? How will you handle disagreements? —SS]